Library construction protocol for nanogram amounts of total RNA for RNAseq – HIGH VOLUME

Jain Group Protocols, GUDMAP Consortium

1) 3ng, 10ng or 30ng total RNA (in 9µl) was reverse transcribed and amplified using the SMARTer® Ultra™ Low Input RNA for Illumina® Sequencing - HV (Clontech Laboratories, cat# 634828) according to the manufacturer’s instructions.

2) The cDNA was sheared for 3 minutes with the Covaris ultra-sonicator.

3) The libraries were prepared using VeraSeq™ ULtra DNA Polymerase (Enzymatics, cat# P7520L) for 12 cycles. PCR conditions were 98°C for 30s; 12 cycles of 98°Cfor 10s, 64°C for 30s, 72°C for 30s; 72°C for 5 min and then hold at 4°C. The forward primer used was: AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTACACGACGCTCTTCCGATCT

The reverse primer with the indexes used were:

Reverse Primer	**Reverse Complement**	**Reported index & Sample Name**
64, CAAGCAGAAGACGGCATACGAGAT**GTAATCA**GTGACTGGAGTTCAGACGTGTGCTCTTCCGA	TGATTAC	TGATTAC-30ng_High-Volume
65, CAAGCAGAAGACGGCATACGAGAT**CCCAGCA**GTGACTGGAGTTCAGACGTGTGCTCTTCCGA	TGCTGGG	TGCTGGG-10ng_High-Volume
66, CAAGCAGAAGACGGCATACGAGAT**CTGTGTC**GTGACTGGAGTTCAGACGTGTGCTCTTCCGA	GACACAG	GACACAG-3ng_High-Volume

4) The purified DNA yield was16.5ng/µl in 30µl and sequencing was done on the Illumina HiSeq2500 following the manufacturer's protocols.